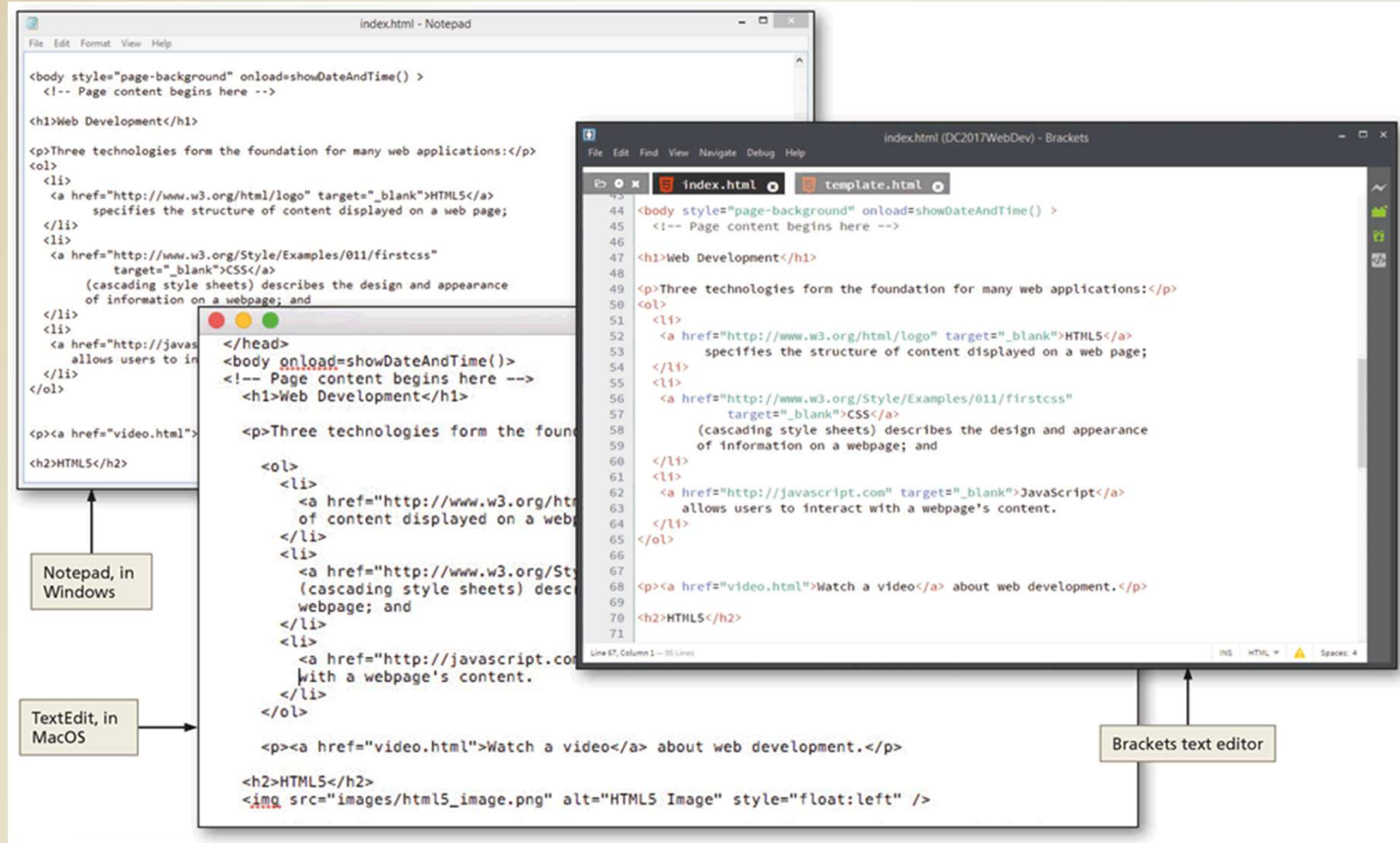


Review Web Technology

Tools for Creating a Website

- Text editors
 - Similar to a word processing program, but lacks most text formatting features
 - Operating systems typically include a text editor
 - A code editor is a type of text editor that contains additional features to help web developers write code

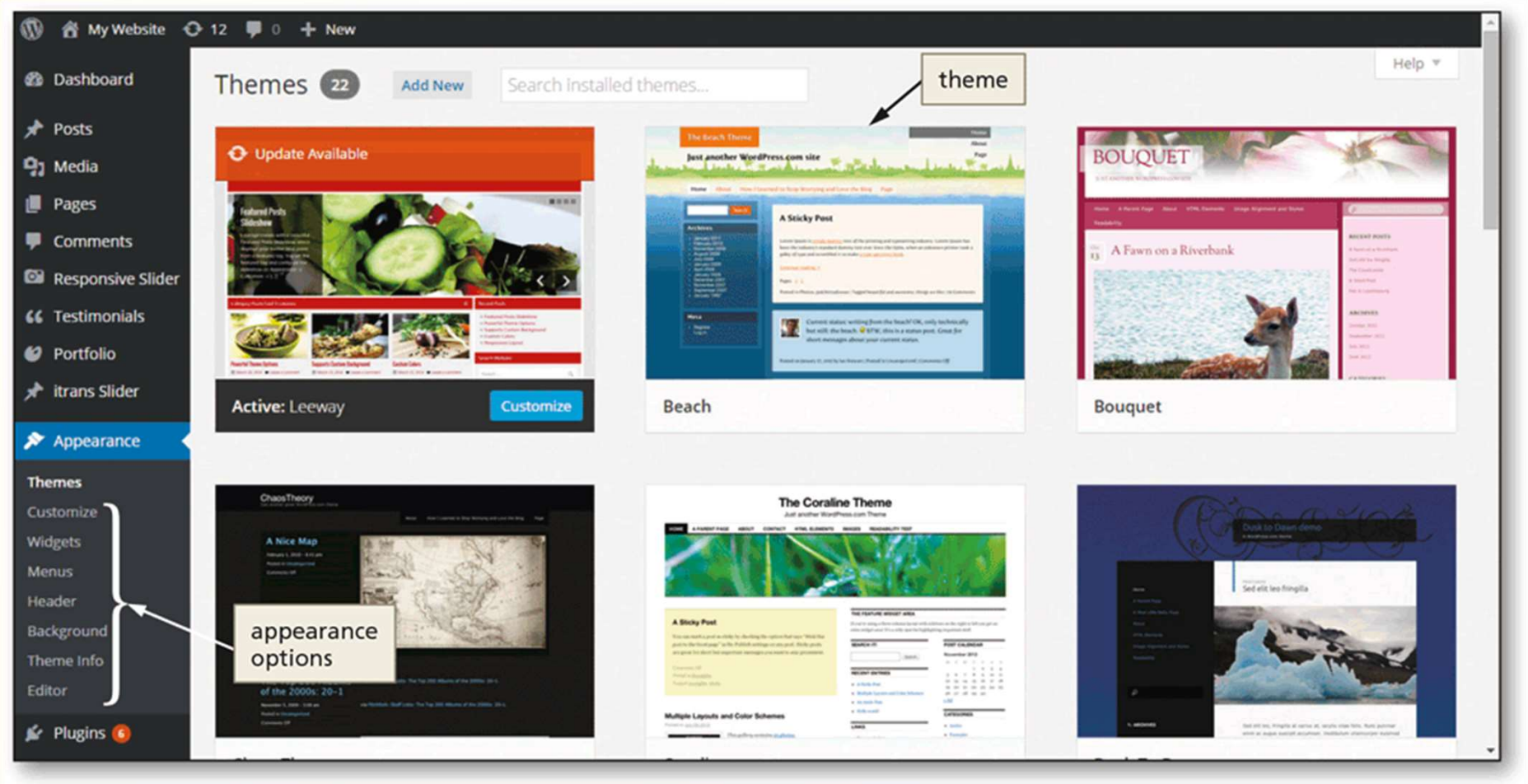
Tools for Creating a Website



Tools for Creating a Website

- Content Management Systems
 - Enables and manages the publishing, modification, organization, and access of various forms of documents and other files on a network or the web
 - Web developer creates a theme, and one or more website content administrators enters the content

Tools for Creating a Website



Website Technologies

- Hypertext Markup Language (HTML) uses a set of codes called tags to format documents for display in a browser
- A complementary technology called cascading style sheets (CSS) contains specifications for the fonts, colors, layout, and placement of these HTML elements on a webpage
- JavaScript is a programming language for creating programs that a browser can run to generate content for a website

Website Technologies

The image shows a screenshot of the Smithsonian National Zoological Park website, specifically the 'Meet Our Animals' page for 'Great Apes & Other Primates'. The website has a green header with navigation links: Visit, Animals, Membership, Science, Education, and Support. A search bar is also present. The main content area features a large heading 'GREAT APES & OTHER PRIMATES' and a sub-heading 'GORILLAS'. Below the gorilla heading, there is a paragraph about six western lowland gorillas. Further down, there is a section titled 'PRIMATES AT THE ZOO' with a photograph of a gorilla. The bottom section is titled 'ABOUT PRIMATES' and contains two paragraphs about the diversity of primates. On the left side, there is a sidebar with links for 'Meet the Primates', 'Facts', 'Exhibit', 'News', 'Science', and 'Enrichment'. At the bottom of the sidebar, there is a fundraising goal: 'Together, We Save Species \$49.50 million'. On the right side, the HTML code for the page is displayed, with annotations pointing to specific elements: 'link to external style sheet' points to the <link> tag in the head; 'h1 header' points to the <h1> tag for 'Great Apes & Other Primates'; 'h2 header' points to the <h2> tag for 'Gorillas'; 'hyperlink' points to the <a> tag in the paragraph about gorillas; 'h3 header' points to the <h3> tag for 'Primates at the Zoo'; 'image' points to the tag for the gorilla photo; and 'paragraph' points to the <p> tag for the text about primates.

link to external style sheet

h1 header

h2 header

hyperlink

h3 header

image

paragraph

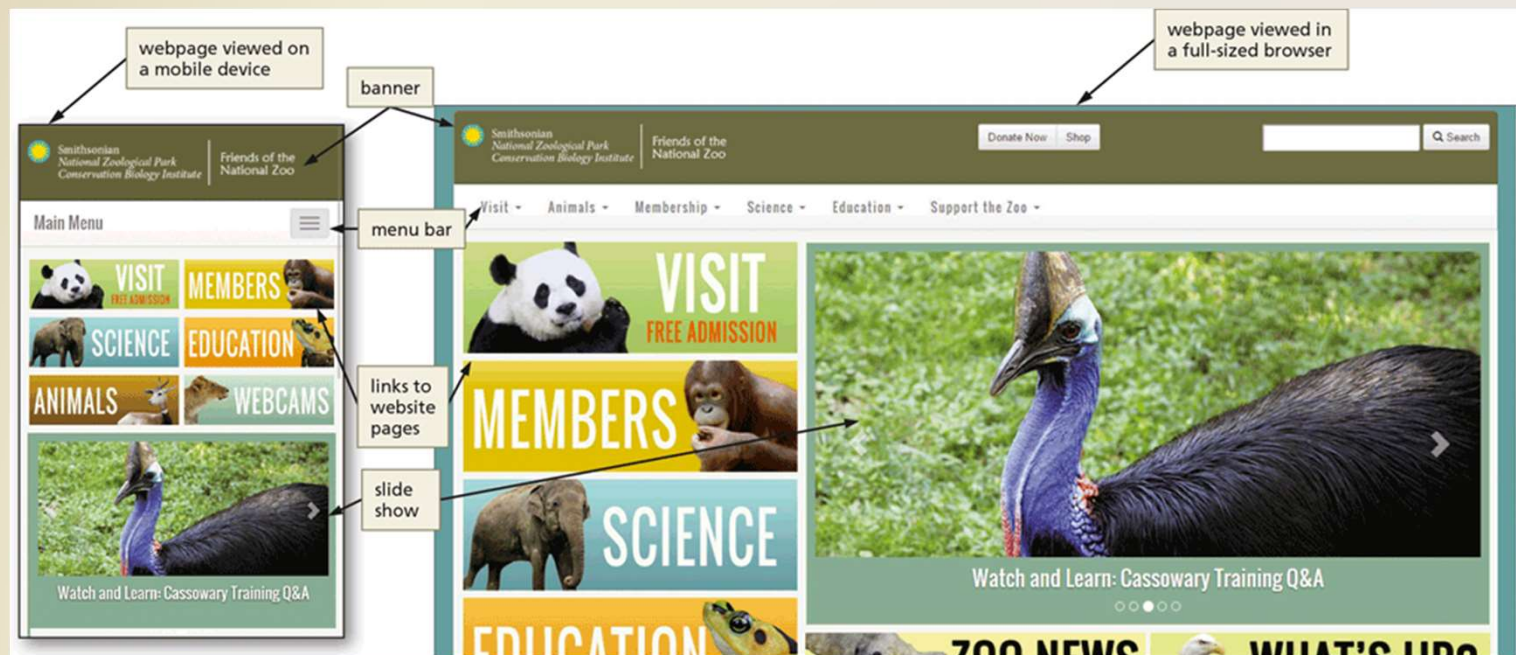
```

1 <!DOCTYPE html
2 PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
3 "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
4 <html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en" lang="en"><!-- Inst
5 codeOutsideHTMLIsLocked="false" -->
6 <head>
7
8 <title>Great Apes and Other Primates - National Zoo</title>
9
10 <link href="https://fonts.googleapis.com/css?family=Oswald:400,300,700" rel=
11 />
12 </head>
13 <body>
14
15 <h1>Great Apes & Other Primates</h1>
16
17 <h2>Gorillas</h2>
18
19 <p>
20 <a href="/Animals/Primates/MeetPrimates/MeetGorillas/">Six western lowland
21 youngest is Kibibi, born in 2009.
22 </p>
23
24 <h3>Primates at the Zoo </h3>
25
26 <a href="http://www.flickr.com/photos/nationalzoo/3707340844/" title="Goril
27 
29 <p>
30 The Zoo is home to many primates. Orangutans and western lowland gorillas
31 tamarins, Geoffroy's marmosets, and howler monkeys, can be found in the <a
32 Gibbon Ridge and lemurs at <a href="/animals/primates/meetprimates/meetlemur
33 src="/admin/images/icons/img_arrowdk.gif" alt="" width="15" height="11" />F
34 </p>
35 <p>
36 On mild days, the orangutans can sometimes be seen overhead as they travel
37 Ape House and Think Tank. </p>
38 <h3>About Primates</h3>
39
40 <p>There are 376 species of primates in the world&#8212;from humans and apes
41 </p>
42 <p>
43 The smallest primate is the pygmy mouse lemur, which can fit in the palm of your
44 pounds. Most primates live in warm climates, and most depend on forests for
45 src="/admin/images/icons/img_arrowdk.gif" alt="" width="15" height="11" />M
46 </p>
47
48 </body>
49 </html>

```

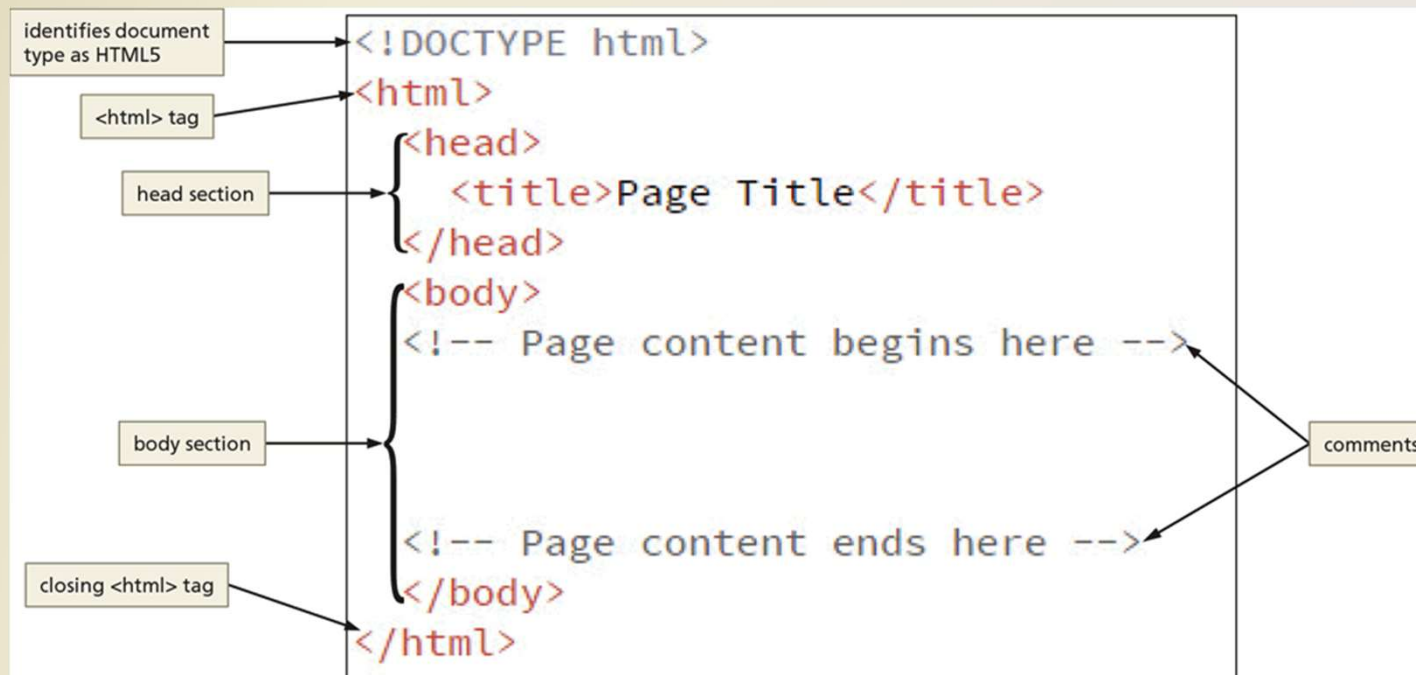
Website Technologies

- Responsive webpages automatically adjust the size of their content to display appropriately relative to the size of the screen of the device on which it is displayed



Structure of a Webpage

- A webpage's source code contains text marked up with HTML tags that instruct a browser how to display that content



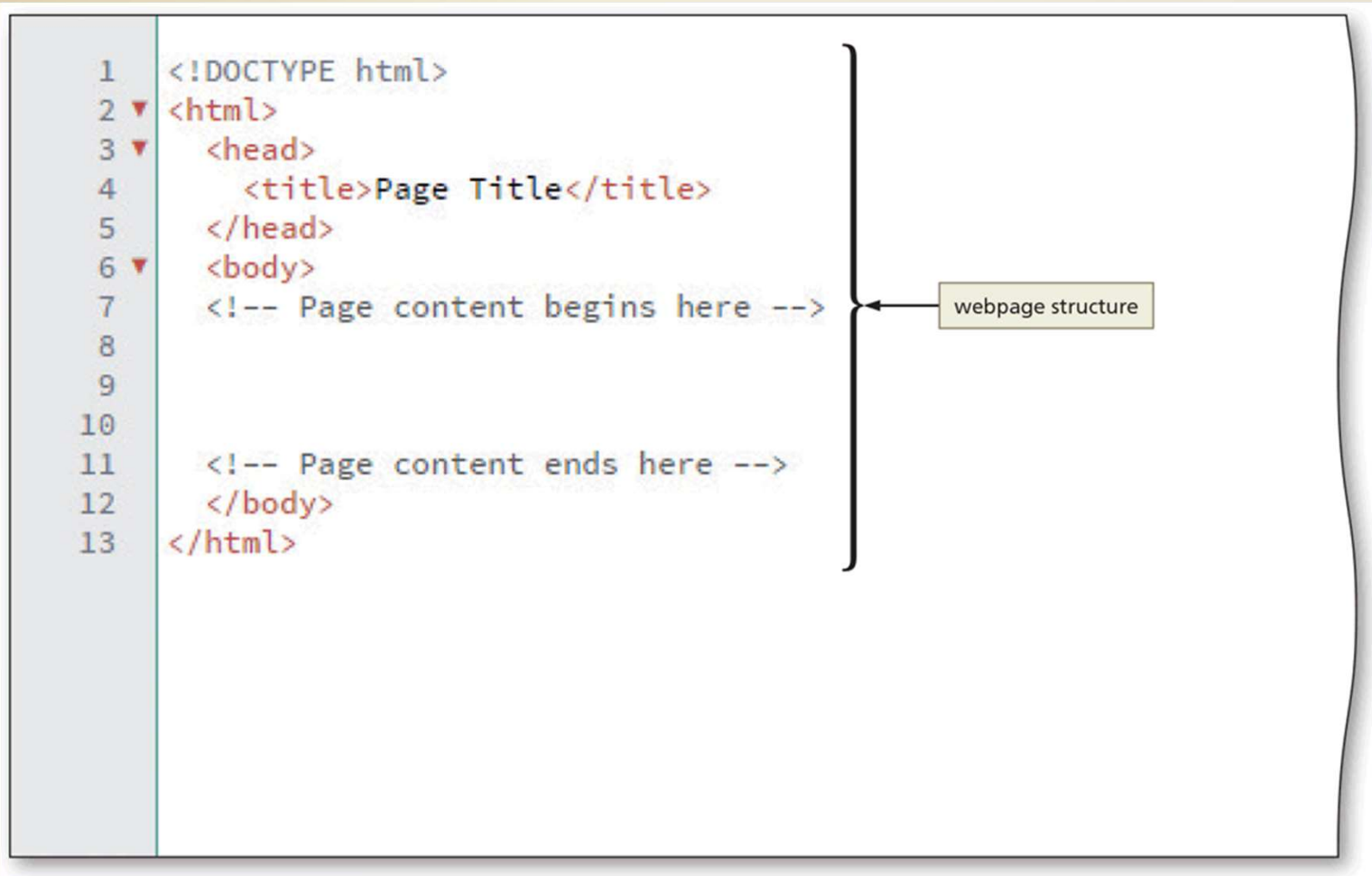
Structure of a Webpage

- The World Wide Web Consortium (W3C) oversees the specification of HTML Standards
- The W3C provides a free, online HTML5 validator application to ensure that a webpage's HTML tags follow the specifications, or rules, for HTML5

Creating the index.html File

- Run the text editor of your choice
- Navigate to and open the template.html file
- If necessary, enable the word wrap feature so that you can view all the webpage text without scrolling horizontally
- Save the file using the file name, index.html. Do not exit the text editor

Creating the index.html File



The diagram shows a code editor window with a light blue sidebar on the left containing line numbers 1 through 13. The main area displays the following HTML code:

```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Page Title</title>
5   </head>
6   <body>
7     <!-- Page content begins here -->
8
9
10
11   <!-- Page content ends here -->
12 </body>
13 </html>
```

A large right-facing curly bracket spans from line 2 to line 13. A yellow rectangular box labeled "webpage structure" has an arrow pointing to the middle of this bracket.

Adding the Webpage Title

- Select the text, Page Title, that appears between the `<title>` and `</title>` tags
- **Type** Mark 's Web Development Page as the title. Replace the name, Mark, with your first name

Headings

- Headings indicate the different sections of a webpage. HTML supports six levels of headings, which are identified by the following tags: `<h1>`, `<h2>`, `<h3>`, `<h4>`, `<h5>`, and `<h6>`

Headings



Paragraphs

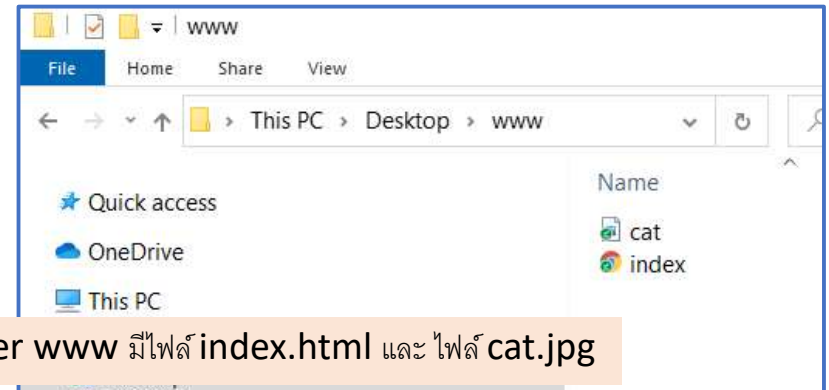
- The `<p>` and `</p>` tags are used to identify the beginning and ending of Paragraphs
- If you have several paragraphs of text on your webpage, these tags will inform the browser to insert additional line spacing above and below the paragraph so that the text is easier to read when displayed in the browser
 - The browser ignores line breaks and line spacing in the HTML file, so it is important to properly define the paragraphs using the `<p>` and `</p>` tags

Images

- Images can be either photos or graphics
- Images always are stored as separate files, and references to the images appear in the HTML code using the `<img>` tag
- Common attributes for the `<img>` tag describe the location of an image file, alternate text for the image, and a style that indicates how to position the image

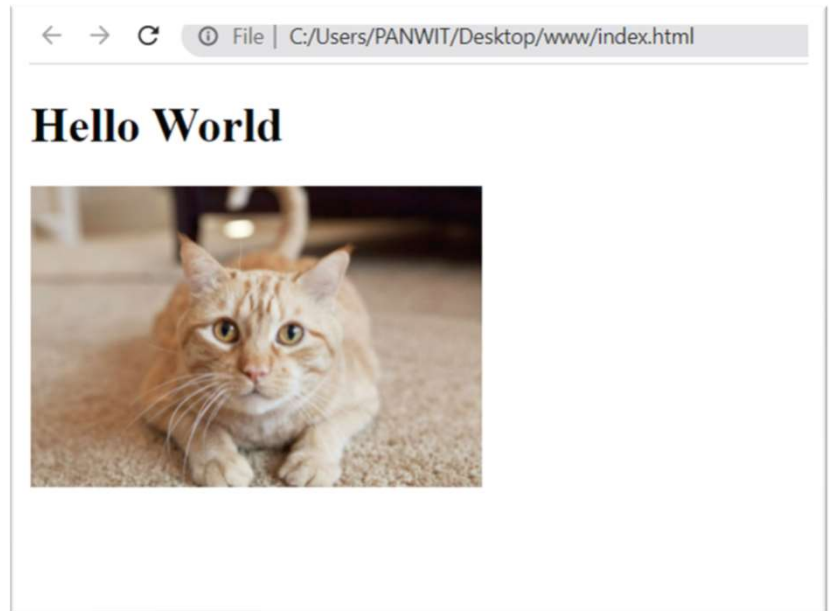
การอ้างอิงตำแหน่งไฟล์รูป ใน html

- File นั้นๆ อยู่ใน folder เดียวกัน
เช่น ไฟล์รูป และไฟล์ index.html
อยู่ใน **folder** เดียวกัน



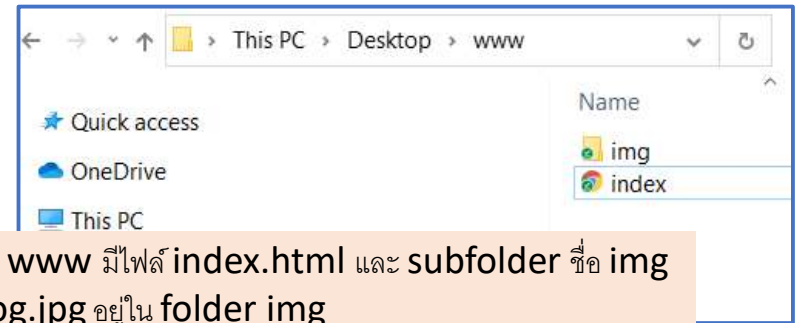
ไฟล์ index.html

```
<!DOCTYPE html>
<html>
  <head>
    <title>My Website Title</title>
  </head>
  <body>
    <h1>Hello World</h1>
    
  </body>
</html>
```



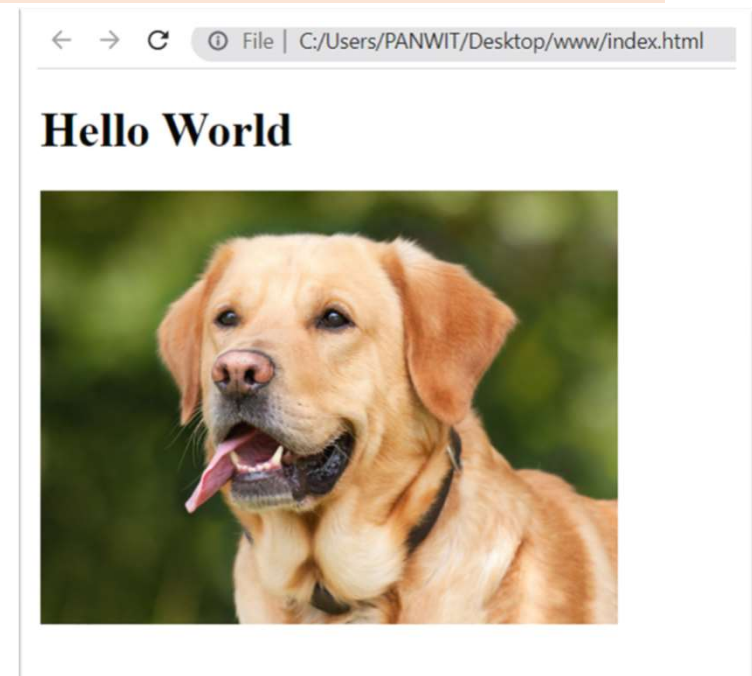
การอ้างอิงตำแหน่งไฟล์รูป ใน html

- File นั้นๆ อยู่คนละ folder เช่น ไฟล์รูป และไฟล์ index.html อยู่คนละ **folder**



ไฟล์ index.html

```
<!DOCTYPE html>
<html>
  <head>
    <title>My Website Title</title>
  </head>
  <body>
    <h1>Hello World</h1>
    
  </body>
</html>
```



Links

- A link, or hyperlink, can be text or an image in a webpage that a user clicks to navigate to another webpage, download a file, or perform another action, such as running an email app and addressing an email message

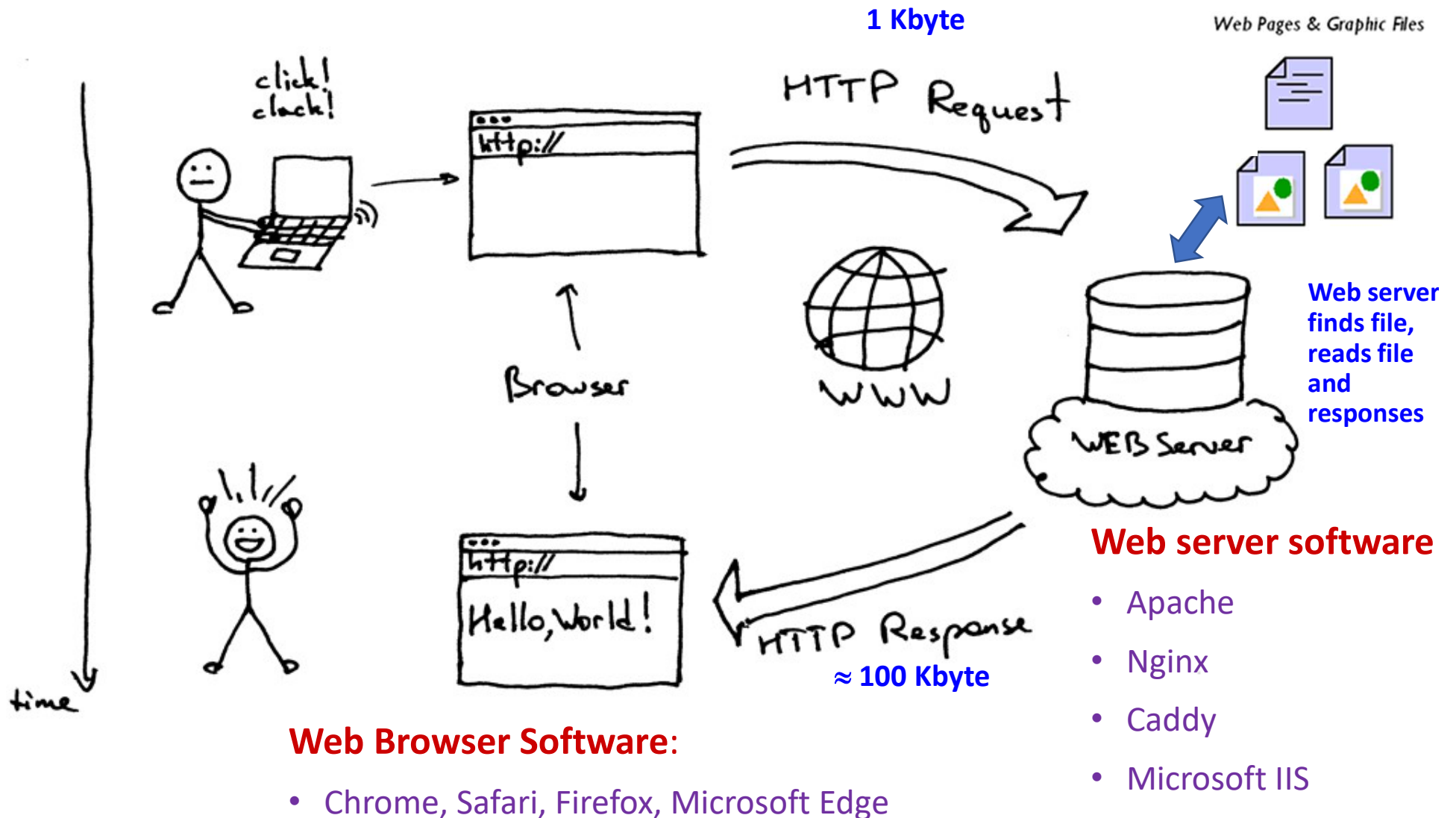
HYPERLINK WITH ABSOLUTE REFERENCE

```
<a href="http://www.w3.org" target="_blank">HTML5</a>
```

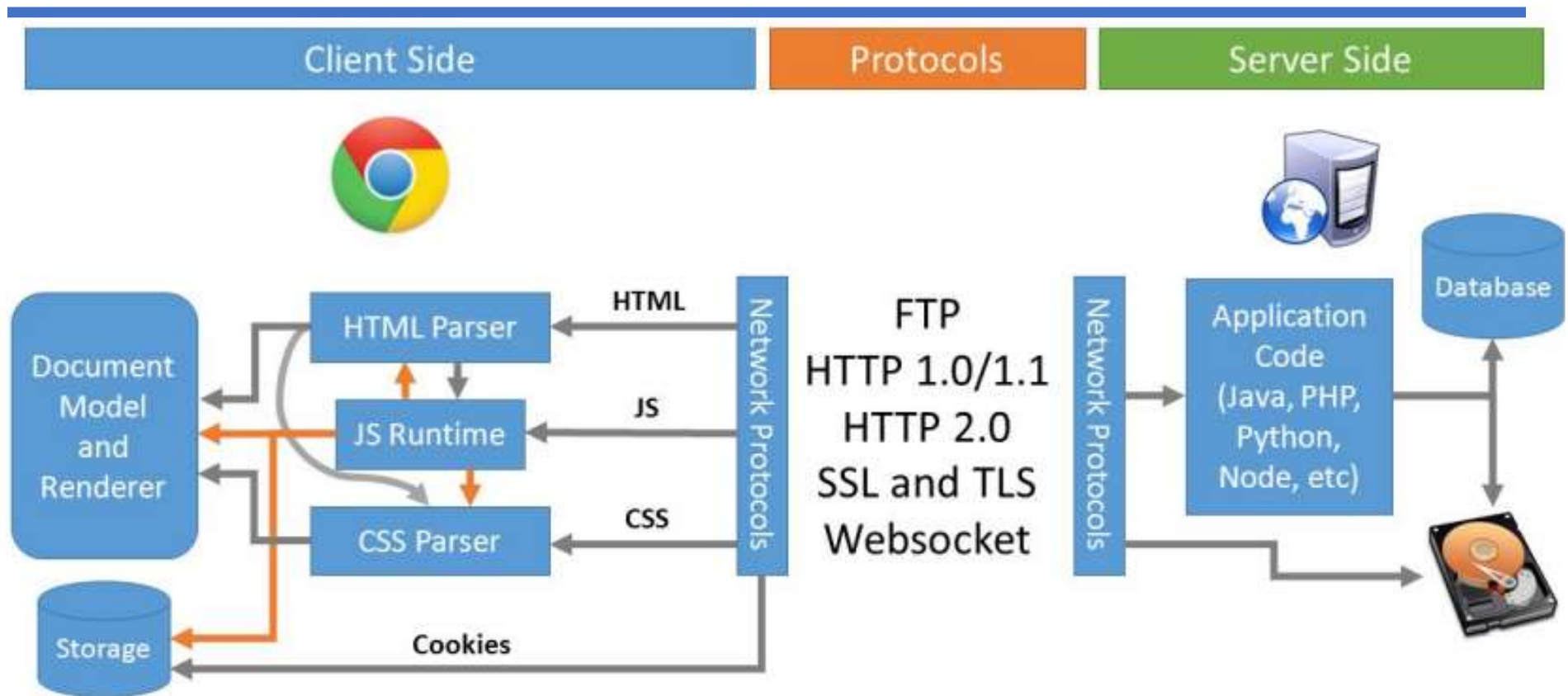
HYPERLINK WITH RELATIVE REFERENCE

```
<a href="video.html"> Watch a video</a>
```

1. ภาพรวม web client-server



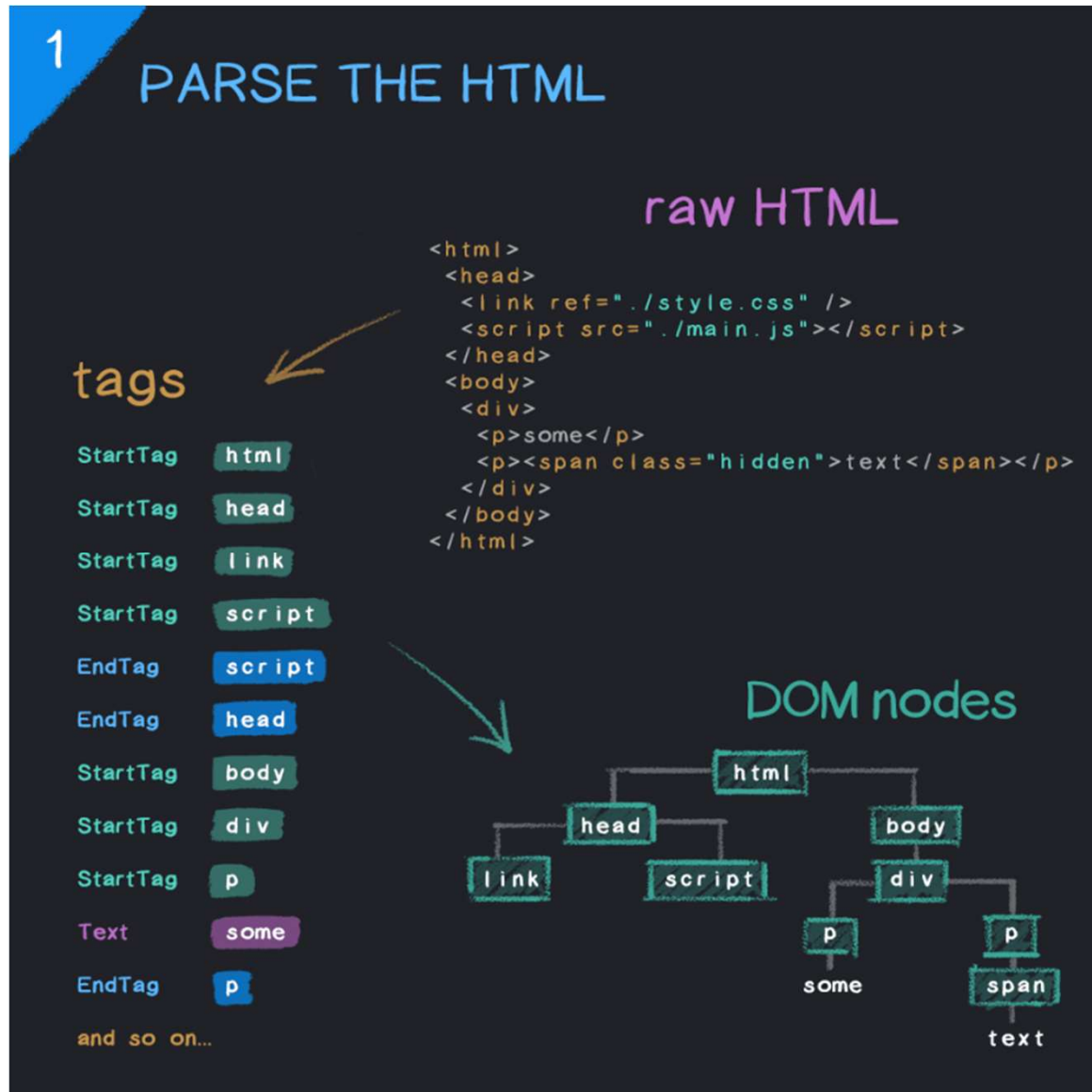
2. การทำงานของ Web Browser



ตัววิเคราะห์คำ แยกคำ ในประโยค

- **HTML Parser** : ทำหน้าที่ render พวก tag ต่าง ๆ ของ code html ออกมาเป็น DOM
 - **CSS Parser** : ส่วนไหนใน code html เป็น css ก็เข้า CSS Parser ออกมาเป็น CSSOM
 - **JS Runtime** : ถ้าใน code html มี JavaScript (js) มันก็จะต้องการ JS runtime environment ในการ run ให้ code js ทำงาน โดยสามารถใช้ js ทำการ อ่าน/แก้ไข html หรือจัดการ css ได้ ซึ่ง browser ทุกตัวบนโลกเดี๋ยวนี้นี้มี js runtime อยู่แล้ว
- JavaScript Runtime ทำหน้าที่เพื่อ อ่าน code ภาษา java script และทำงานตามคำสั่ง เพื่อให้ OS ต่างๆ ทำงานได้

2. การทำงานของ Web Browser



2. การทำงานของ Web Browser

2

FETCH EXTERNAL RESOURCES

CSS blocks rendering

JavaScript blocks parser...

unless!

defer waits until parser is finished

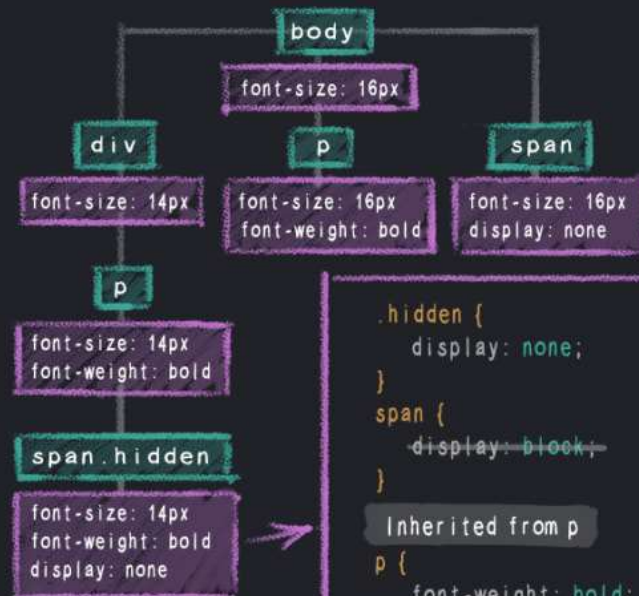
async executes as soon as it loads

2. การทำงานของ Web Browser

3

PARSE THE CSS AND BUILD THE CSSOM

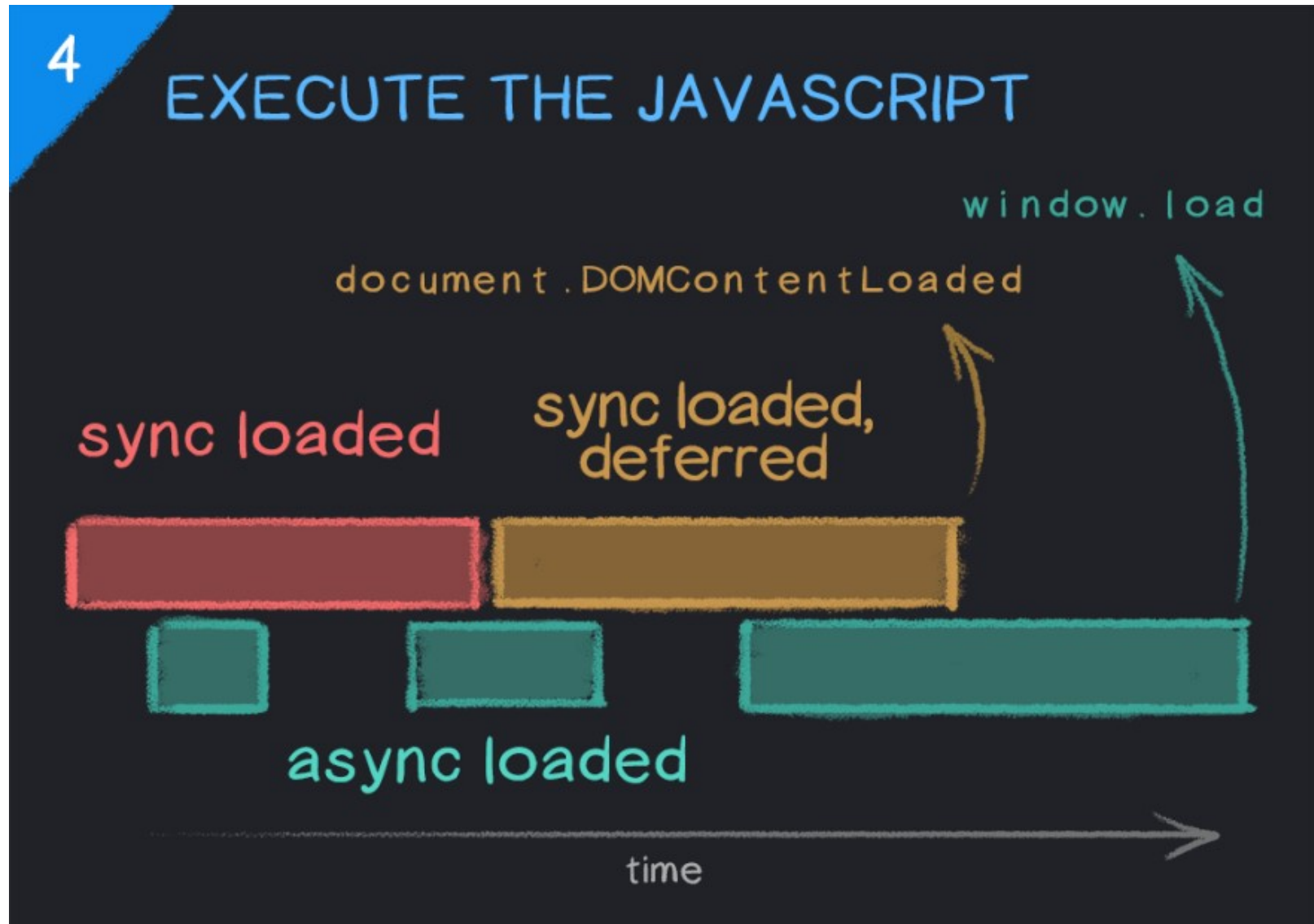
```
body {  
  font-size: 16px;  
}  
  
div {  
  font-size: 14px;  
}  
  
p {  
  font-weight: bold;  
}  
  
.hidden {  
  display: none;  
}  
  
span {  
  display: block;  
}
```



Any CSS style can potentially be overridden or changed by some other style loaded on the page, which is why the CSSOM cannot be fully constructed until all of the page's stylesheets have been loaded.

```
.hidden {  
  display: none;  
}  
span {  
  display: block;  
}  
Inherited from p  
p {  
  font-weight: bold;  
}  
Inherited from div  
div {  
  font-size: 14px;  
}  
Inherited from body  
body {  
  font-size: 16px;  
}
```

2. การทำงานของ Web Browser

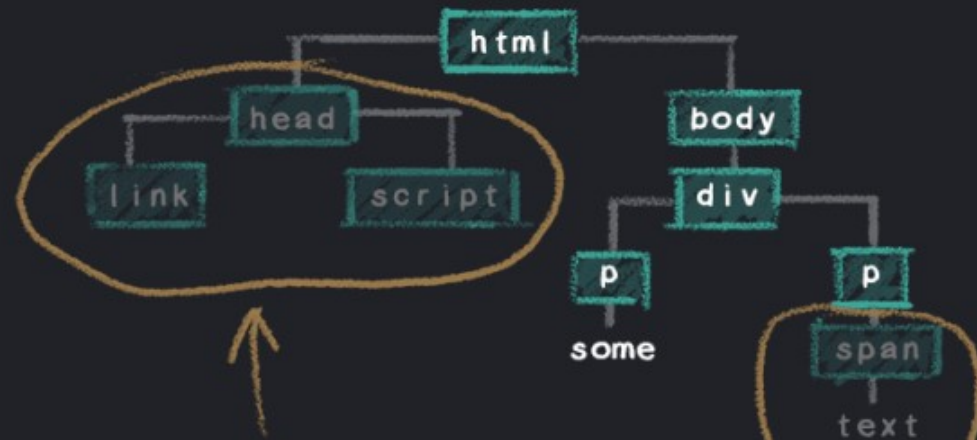


2. การทำงานของ Web Browser

5

MERGE DOM AND CSSOM TO CONSTRUCT THE RENDER TREE

```
body {  
  font-size: 16px;  
}  
  
div {  
  font-size: 14px;  
}  
  
p {  
  font-weight: bold;  
}  
  
span {  
  display: none;  
}
```



non-visible nodes

excluded from the render tree

"display: none"

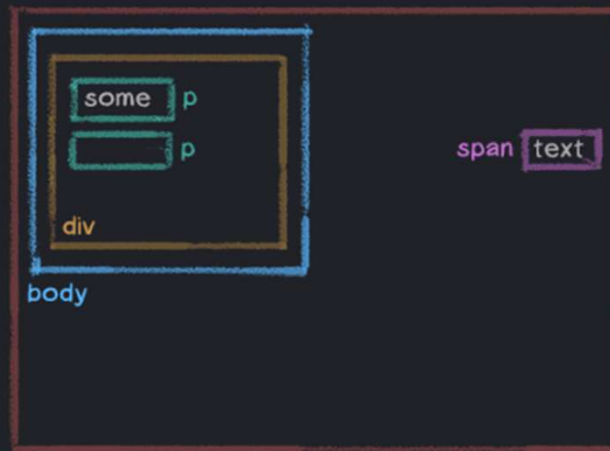
2. การทำงานของ Web Browser

6

CALCULATE LAYOUT AND PAINT

```
<html>
<head>
  <link ref="./style.css" />
</head>
<body>
  <div>
    <p>some</p>
    <p><span>text</span></p>
  </div>
</body>
</html>
```

html



let's add some layout styles...

```
* {
  padding: 25px;
}

html {
  height: 100%;
}

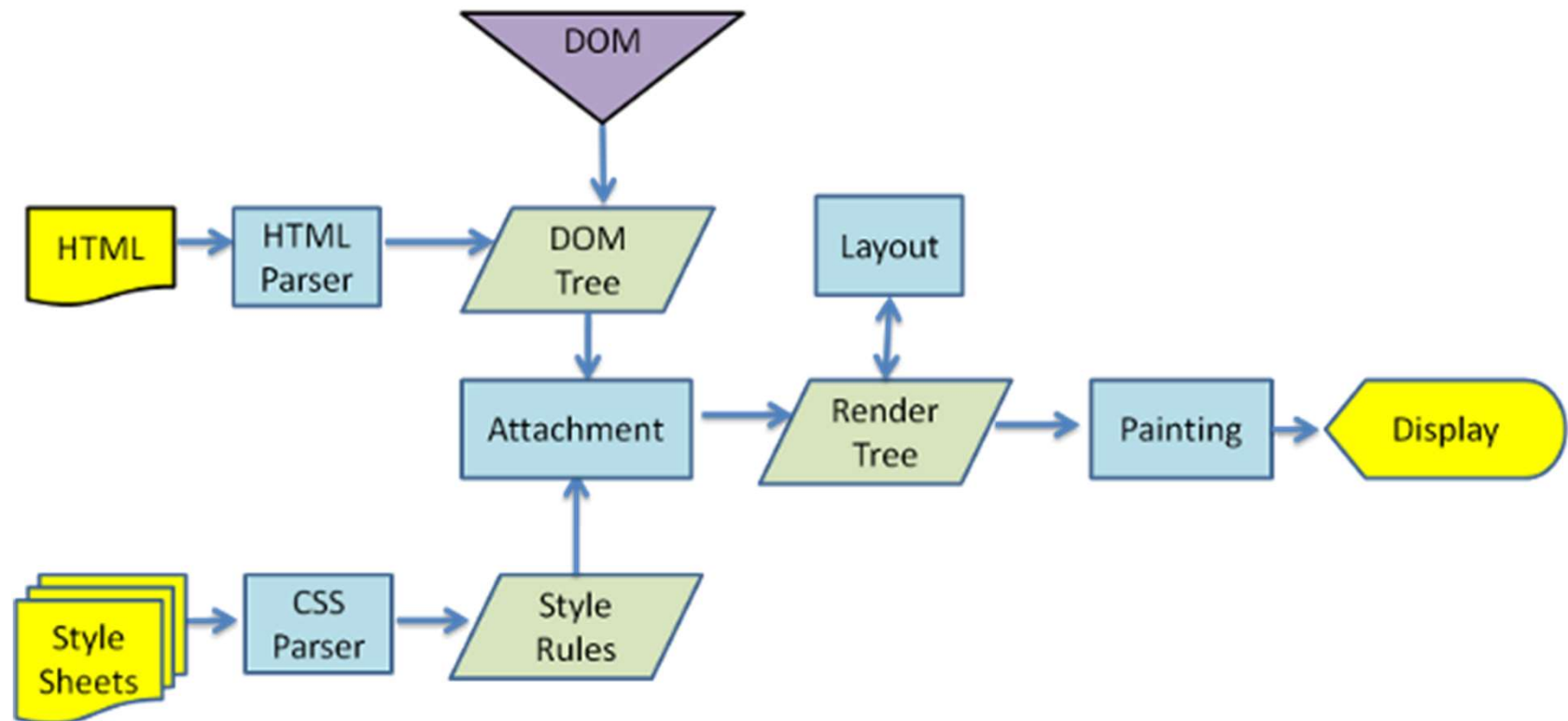
body {
  width: 50%;
}

div {
  height: 200px;
}

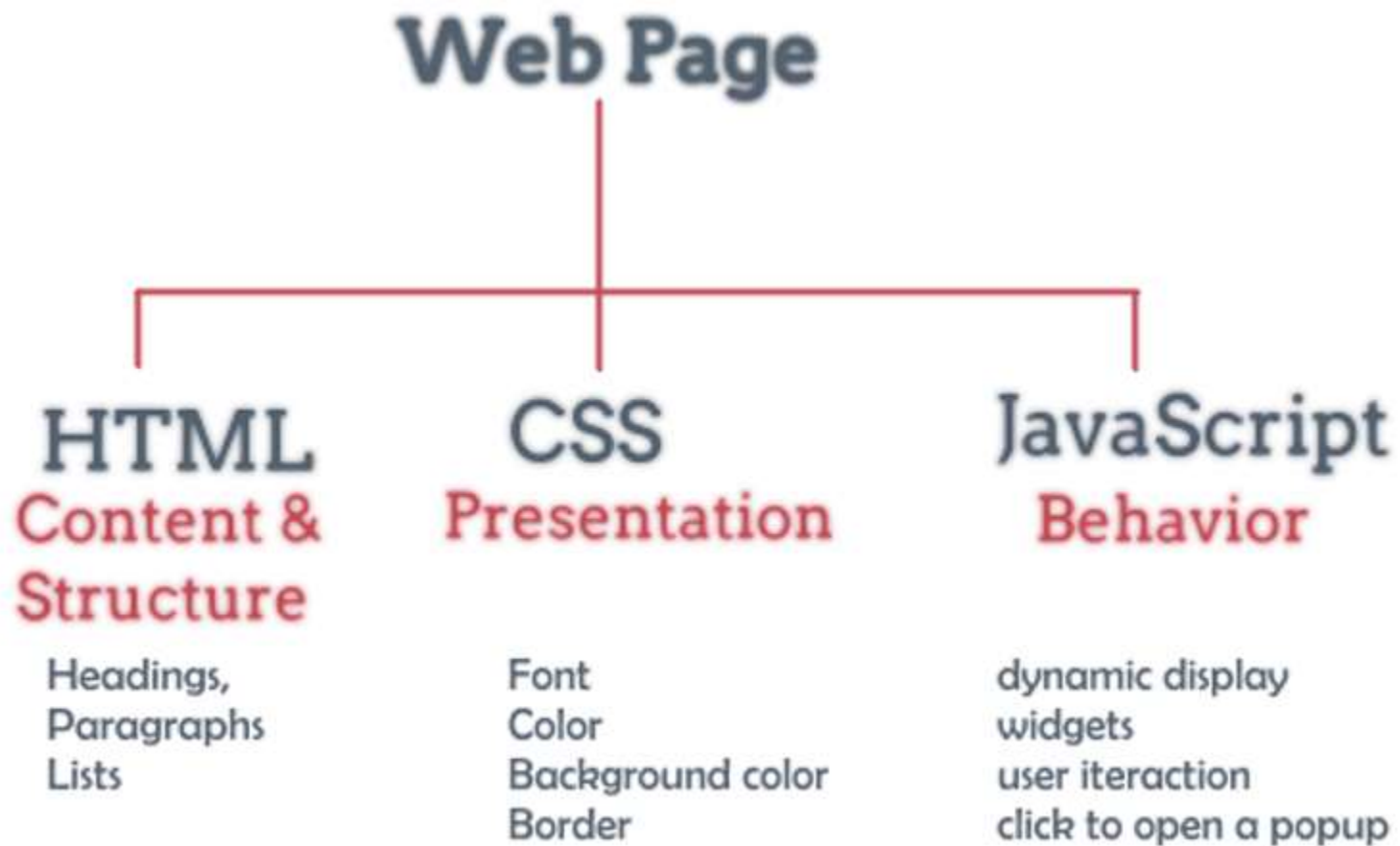
p {
  width: 50%;
}

span {
  position: absolute;
  right: 25px;
}
```

2. การทำงานของ Web Browser

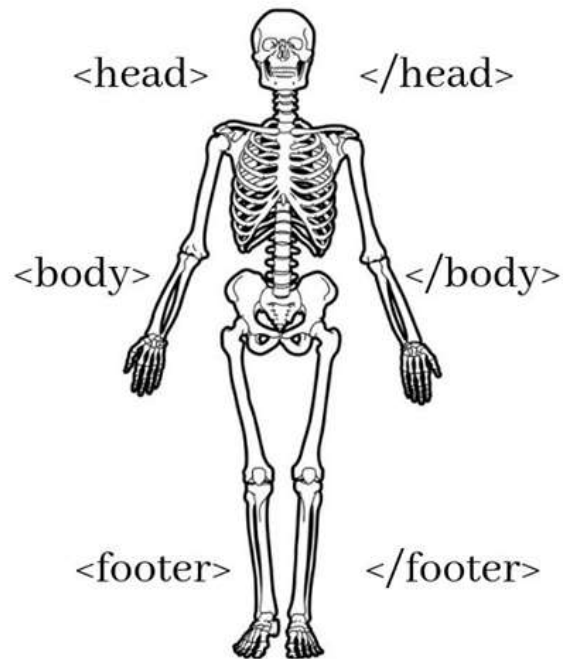


3. Html CSS JavaScript



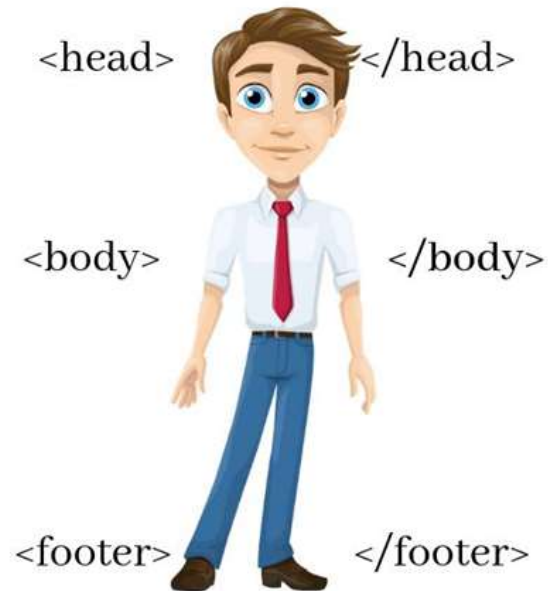
HTML Vs CSS

HTML



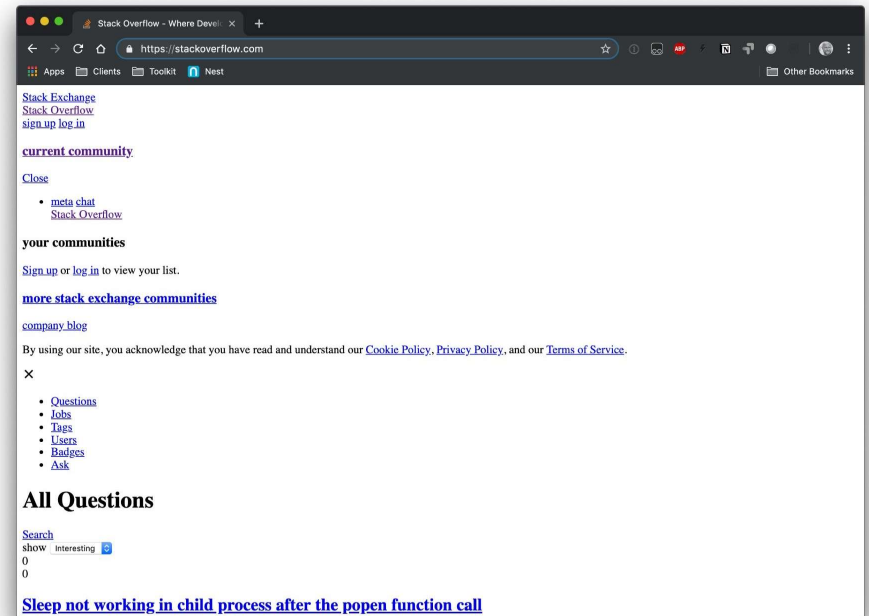
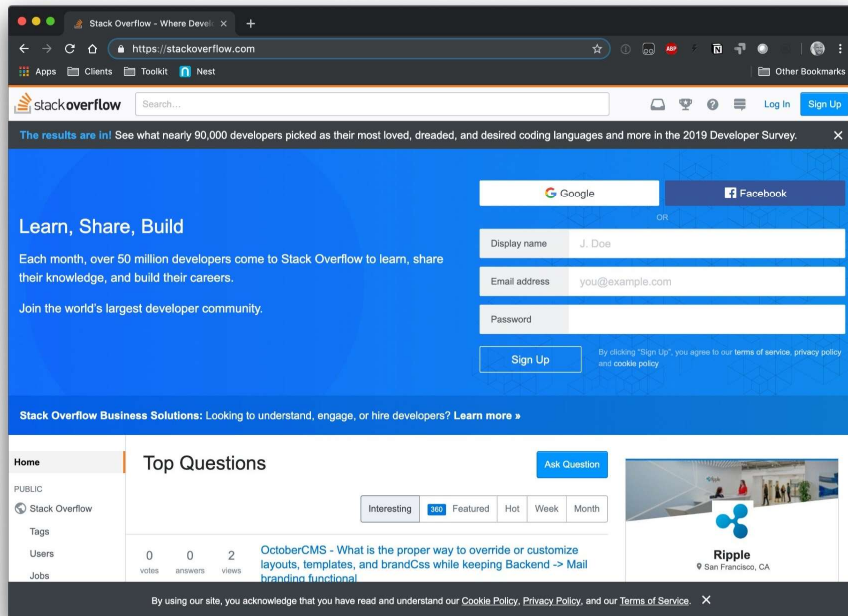
Structural Layer

HTML + CSS



Presentational Layer

HTML Vs CSS

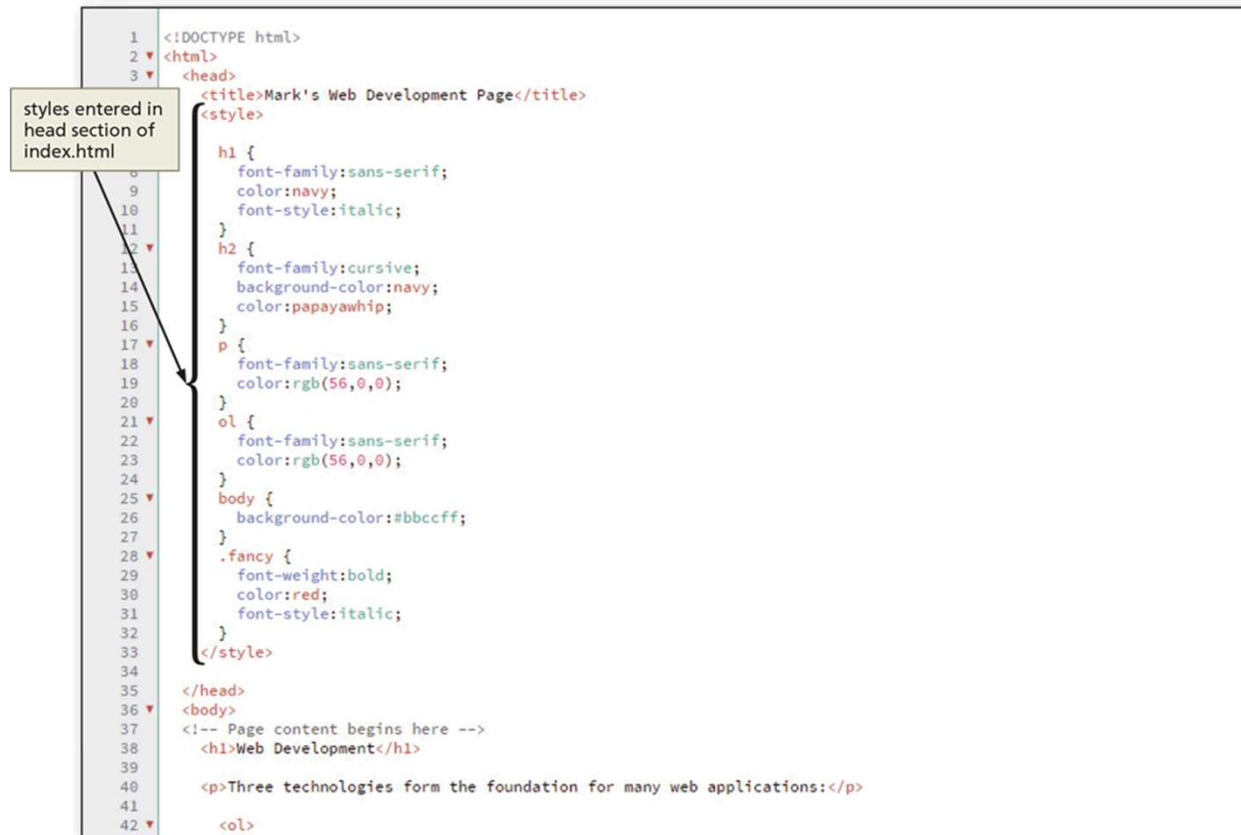


3. Applying Styles with CSS

- Although it is possible to customize these elements of webpages using HTML, CSS makes it easier to specify the appearance of similar elements in the same webpage or same website
- Embedded styles, which **define styles in the <head> section of the index.html document, apply to the entire webpage** on which they are defined

```
<style>
  h1 {
    font-family:sans-serif;
    color:navy;
    font-style:italic;
  }
  h2 {
    font-family:cursive;
    background-color: navy;
    color:papayawhip;
  }
  p {
    font-family:sans-serif;
    color:rgb(56,0,0);
  }
  ol {
    font-family:sans-serif;
    color:rgb(56,0,0);
  }
  body {
    background-color:#bbccff;
  }
  .fancy {
    font-weight:bold;
    color:red;
    font-style:italic;
  }
</style>
```

Adding CSS to the index.html File

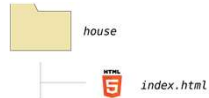


```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Mark's Web Development Page</title>
5   <style>
6     h1 {
7       font-family:sans-serif;
8       color:navy;
9       font-style:italic;
10    }
11    h2 {
12      font-family:cursive;
13      background-color:navy;
14      color:papayawhip;
15    }
16    p {
17      font-family:sans-serif;
18      color:rgb(56,0,0);
19    }
20    ol {
21      font-family:sans-serif;
22      color:rgb(56,0,0);
23    }
24    body {
25      background-color:#bbccff;
26    }
27    .fancy {
28      font-weight:bold;
29      color:red;
30      font-style:italic;
31    }
32  </style>
33 </head>
34 <body>
35   <!-- Page content begins here -->
36   <h1>Web Development</h1>
37   <p>Three technologies form the foundation for many web applications:</p>
38   <ol>
```

styles entered in head section of index.html

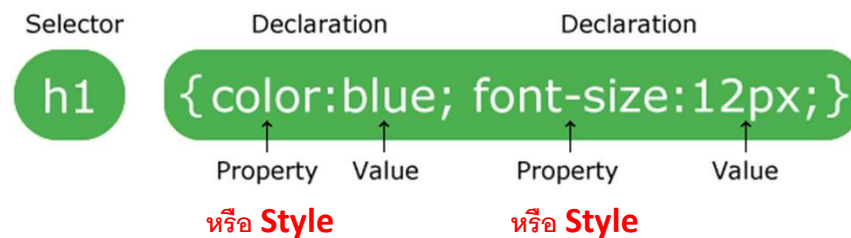
Styling HTML with CSS

- CSS can be added to HTML elements in 3 ways
 - Inline - by using the style attribute in HTML elements
 - Internal - by using a <style> element in the <head> section
 - External - by using an external CSS file

Inline	Internal	External
<pre><h1 style="color:blue;">This is a Blue Heading</h1></pre>	<pre><!DOCTYPE html> <html> <head> <style> body {background-color: powderblue;} h1 {color: blue;} p {color: red;} </style> </head> <body> <h1>This is a heading</h1> <p>This is a paragraph.</p> </body> </html></pre>	<pre><!DOCTYPE html> <html> <head> <link rel="stylesheet" href="styles.css"> </head> <body> <h1>This is a heading</h1> <p>This is a paragraph.</p> </body> </html></pre>
		

3. CSS Syntax

A CSS rule-set consists of a selector and a declaration block:



- The **selector** points to the **HTML element** you want to style.
- The declaration block contains one or more declarations separated by semicolons.
- Each declaration includes a CSS **property name** and a **value**, separated by a colon.
- A CSS declaration always ends with a semicolon, and declaration blocks are surrounded by curly braces.

ที่มา https://www.w3schools.com/css/css_syntax.asp

4. JavaScript



Html CSS JavaScript



HTML



HTML + CSS



**HTML + CSS
+ JAVASCRIPT**



house folder



index.html



styles.css



scripts.js

4. JavaScript

- เดิมทีภาษา **JavaScript** ทำงานที่ **Web browser** เป็นหลัก โดยทำหน้าที่เปลี่ยนแปลงโครงสร้างของ **HTML** เป็นหลัก



- JavaScript Can Change HTML Content
- JavaScript Can Change HTML Attribute Values
- JavaScript Can Change HTML Styles (CSS)
- JavaScript Can Hide/Show HTML Elements

4. JavaScript

- **JavaScript Can Change HTML Content**

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2>What Can JavaScript Do?</h2>
```

```
<p id="demo">JavaScript can change HTML content.</p>
```

```
<button type="button" onclick="document.getElementById('demo').innerHTML  
= 'Hello JavaScript!'">Click Me!</button>
```

```
</body>
```

```
</html>
```

What Can JavaScript Do?

JavaScript can change HTML content.

Click Me!

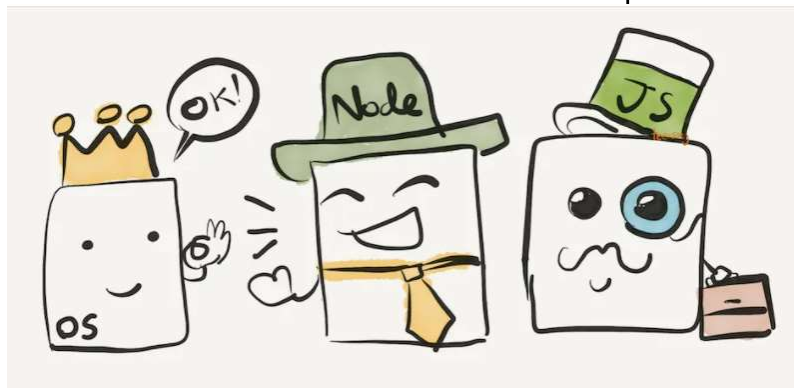
What Can JavaScript Do?

Hello JavaScript!

Click Me!

4. JavaScript

- ต่อมา ผู้คนอยากให้ **JavaScript** ทำงานนอก **Web Browser** ได้ เพื่อเพิ่มความสามารถของมัน จึงมีคนคิดค้นสร้างสิ่งที่เรียกว่า **Node**
- **Node** หรือ **Node.js** คือ **JavaScript Runtime** ที่ถูกสร้างด้วย **Chrome'V8 JavaScript Engine**
- โดย **JavaScript Runtime** ทำหน้าที่เพื่อ อ่าน **code** ภาษา **java script** และทำงานตามคำสั่ง เพื่อให้ **os** ต่างๆ ทำงานได้

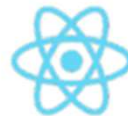


Full Stack JavaScript Tools and Technologies

Front End



Or



React



Knockout.



BACKBONE.JS

node.js +



Back End

Or

METEOR

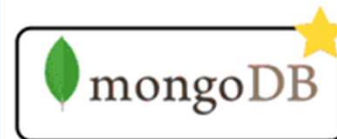
sails

restify



KeystoneJS

Database



Or

MySQL

PostgreSQL

CouchDB

cassandra

5. Introduction to API

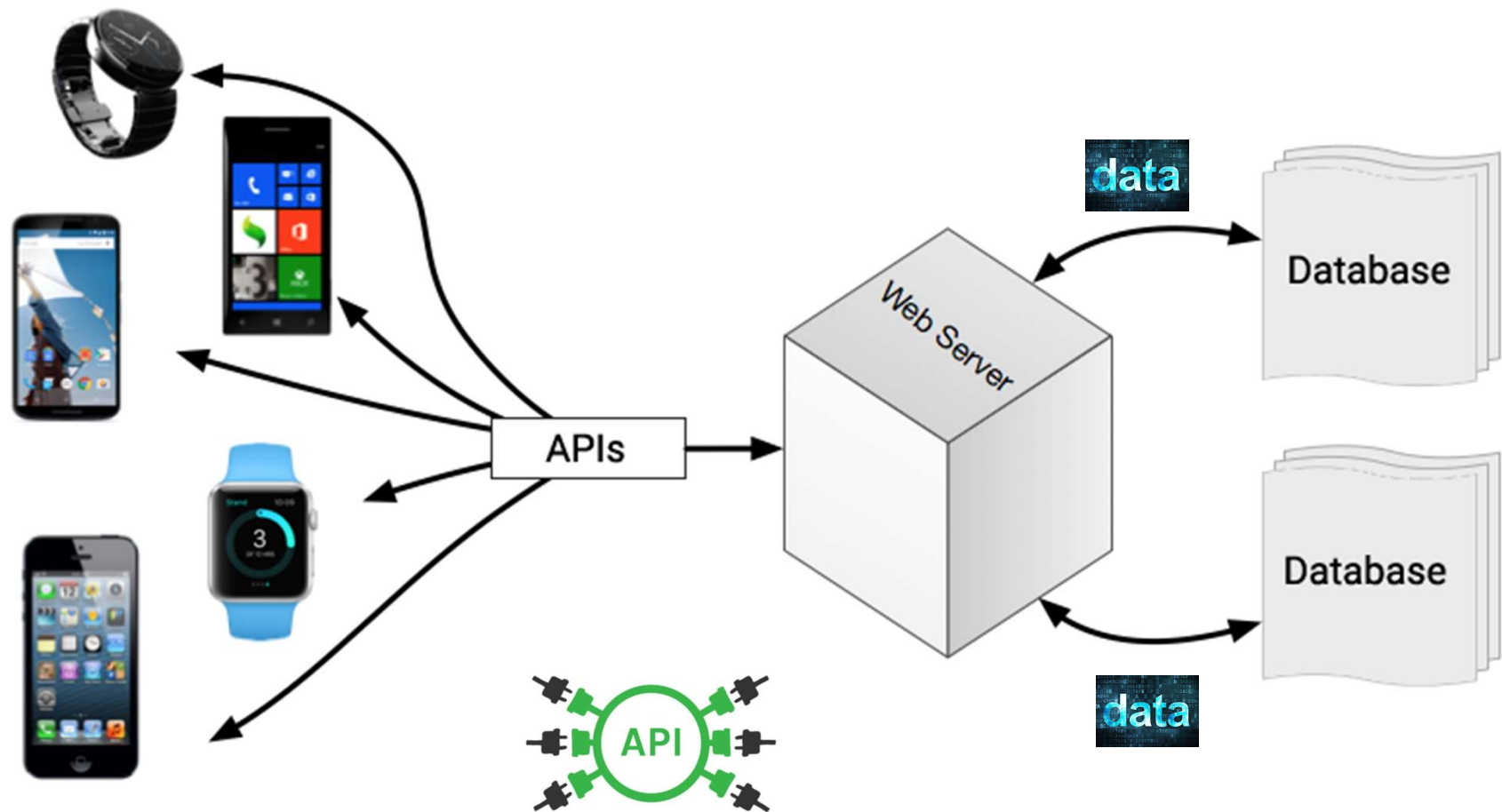


โรงไฟฟ้า ผลิตไฟฟ้า เพื่อให้บริการไฟฟ้า
คำถาม ถ้าไม่มีปลั๊กตัวเมียมาให้ เราจะใช้ไฟฟ้า อย่างไร ?



5. Introduction to API

API ย่อมาจาก Application Programming Interface



An API allows one piece of software talk to another.



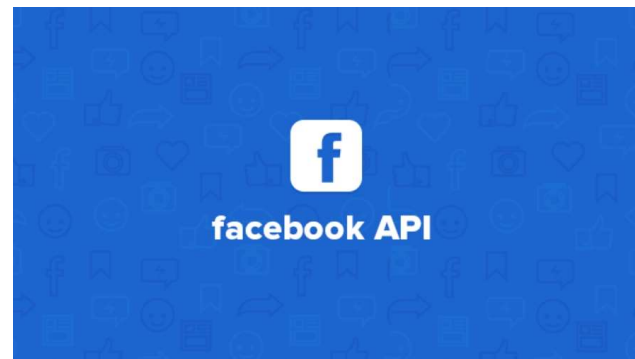
ตัวอย่าง API

Google Nearby API



Google Maps APIs

You Tube
Data API v3



ตัวอย่าง API ให้ข้อมูล อัตราแลกเปลี่ยน

<https://api.exchangeratesapi.io/latest>

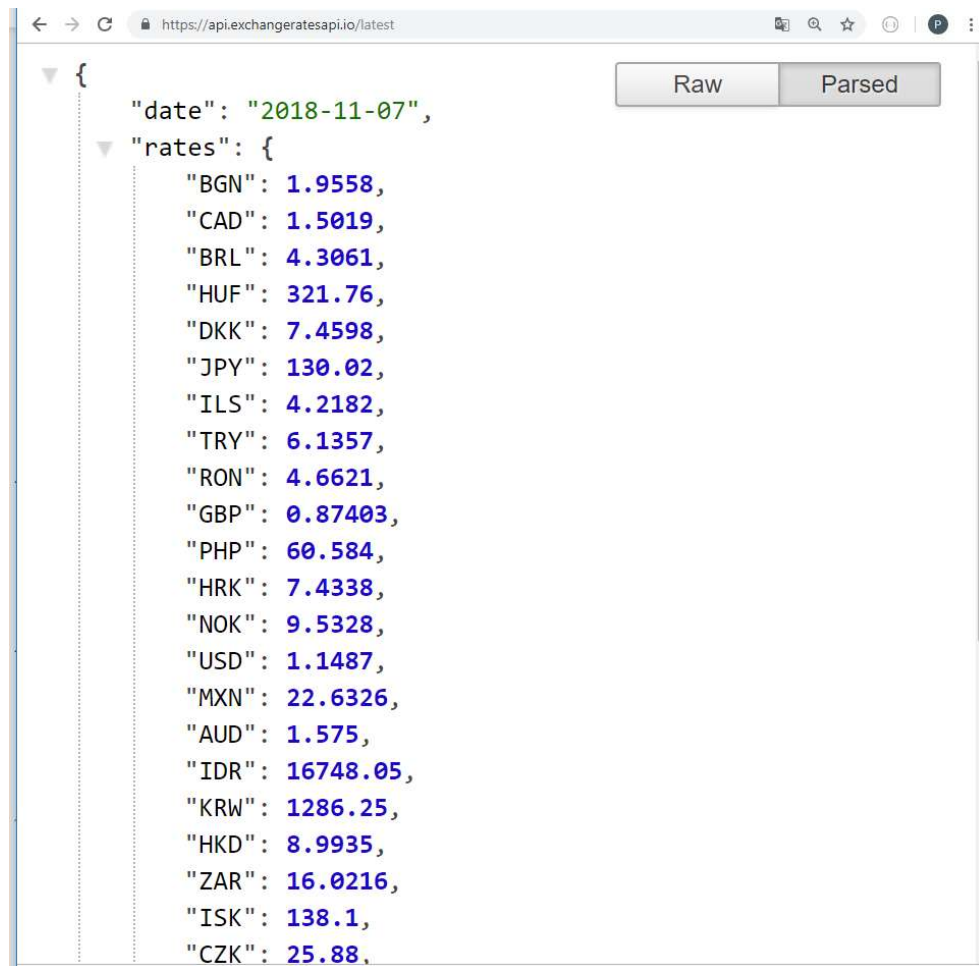


```
{
  "date": "2018-11-07",
  "rates": {
    "BGN": 1.9558,
    "CAD": 1.5019,
    "BRL": 4.3061,
    "HUF": 321.76,
    "DKK": 7.4598,
    "JPY": 130.02,
    "ILS": 4.2182,
    "TRY": 6.1357,
    "RON": 4.6621,
    "GBP": 0.87403,
    "PHP": 60.584,
    "HRK": 7.4338,
    "NOK": 9.5328,
    "USD": 1.1487,
    "MXN": 22.6326,
    "AUD": 1.575,
    "IDR": 16748.05,
    "KRW": 1286.25,
    "HKD": 8.9935,
    "ZAR": 16.0216,
    "ISK": 138.1,
    "CZK": 25.88,
    "THB": 37.66,
    "MYR": 4.7734,
    "NZD": 1.6954,
    "PLN": 4.294,
    "SEK": 10.3313,
    "RUB": 75.7709,
    "CNY": 7.9426,
    "SGD": 1.5729,
    "CHF": 1.1444,
    "INR": 83.28
  },
  "base": "EUR"
}
```

ตัวอย่าง API ให้ข้อมูล อัตราแลกเปลี่ยน

<https://api.exchangeratesapi.io/latest>

ติดตั้ง plugin **json formatter**



The screenshot shows a web browser window with the URL <https://api.exchangeratesapi.io/latest>. The response is displayed as a JSON object, which has been formatted using a 'json formatter' plugin. The JSON structure includes a 'date' field and a 'rates' object containing various currency exchange rates. The 'Raw' and 'Parsed' tabs are visible at the top right of the JSON viewer.

```
{
  "date": "2018-11-07",
  "rates": {
    "BGN": 1.9558,
    "CAD": 1.5019,
    "BRL": 4.3061,
    "HUF": 321.76,
    "DKK": 7.4598,
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    "IDR": 16748.05,
    "KRW": 1286.25,
    "HKD": 8.9935,
    "ZAR": 16.0216,
    "ISK": 138.1,
    "CZK": 25.88,
  }
}
```

การเขียน html ดึงค่าข้อมูล ผ่าน API

<https://api.exchangeratesapi.io/latest>

ไฟล์ index.html

```
File Edit Format View Help
<html>
<head>
  <title>Welcome to JSON</title>
  <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.3.1/jquery.min.js"></script>
</head>
<body>

  <script type="text/javascript">
    $.getJSON('https://api.exchangeratesapi.io/latest', function(response) {
      document.getElementById("data").innerHTML = response.rates.THB
      document.getElementById("data1").innerHTML = response.date
    });
  </script>

  <div style = "color:red" id="data" > </div>
  <div style = "color:blue" id="data1"> </div>

</body>
</html>
```

Request

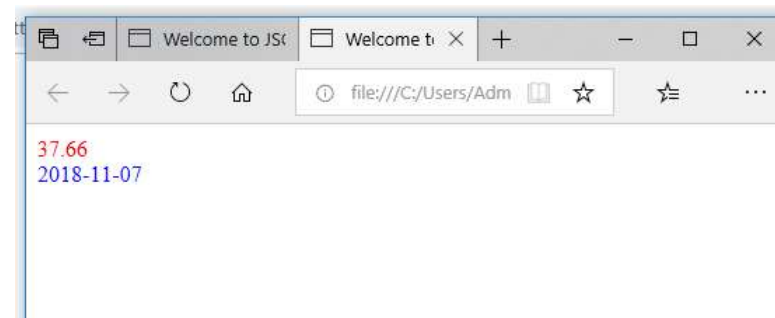
Data

```
https://api.exchangeratesapi.io/latest
{
  "PHP": 60.584,
  "HRK": 7.4338,
  "NOK": 9.5328,
  "USD": 1.1487,
  "MXN": 22.6326,
  "AUD": 1.575,
  "IDR": 16748.05,
  "KRW": 1286.25,
  "HKD": 8.9935,
  "ZAR": 16.0216,
  "ISK": 138.1,
  "CZK": 25.88,
  "THB": 37.66,
  "MYR": 4.7734,
  "NZD": 1.6954,
  "PLN": 4.294,
  "SEK": 10.3313,
  "RUB": 75.7709,
  "CNY": 7.9426,
  "SGD": 1.5729,
  "CHF": 1.1444,
  "INR": 83.28
},
"base": "EUR"
```

เปิดไฟล์ผ่าน Web browser



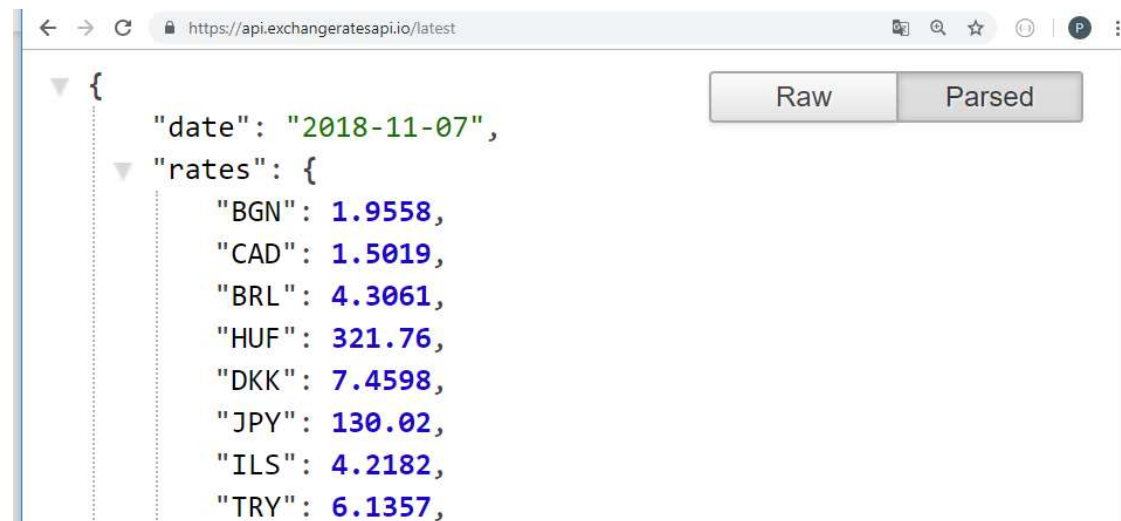
แสดงผล



REST API

- **REST** หรือ Representational State Transfer มันเป็นวิธีในการสร้าง Web Service รูปแบบหนึ่งที่อาศัย HTTP Method (GET, POST, PUT, DELETE) ในการทำงาน และส่งผลกลับมาในรูปแบบของ JSON หรือ XML

เช่น <https://api.exchangeratesapi.io/latest>



REST API

ตัวอย่าง API Document โดยใช้ Swagger

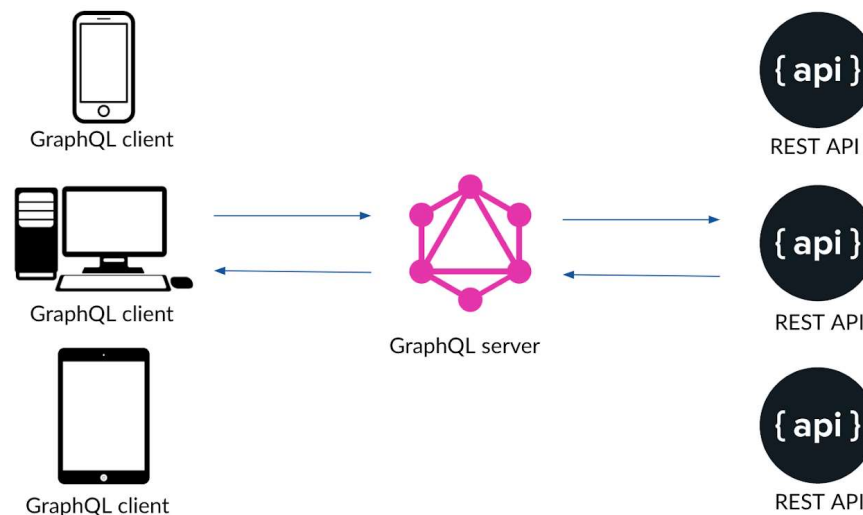
workshop\$Order Entity CRUD operations

**GET****/entities/workshop\$Order** Gets a list of entities: workshop\$Order**POST****/entities/workshop\$Order** Creates new entity: workshop\$Order**GET****/entities/workshop\$Order/{entityId}** Gets a single entity by identifier: workshop\$Order**PUT****/entities/workshop\$Order/{entityId}** Updates the entity: workshop\$Order**DELETE****/entities/workshop\$Order/{entityId}** Deletes the entity: workshop\$Order**GET****/entities/workshop\$Order/search** Find entities by filter conditions: workshop\$Order**POST****/entities/workshop\$Order/search** Find entities by filter conditions: workshop\$Order

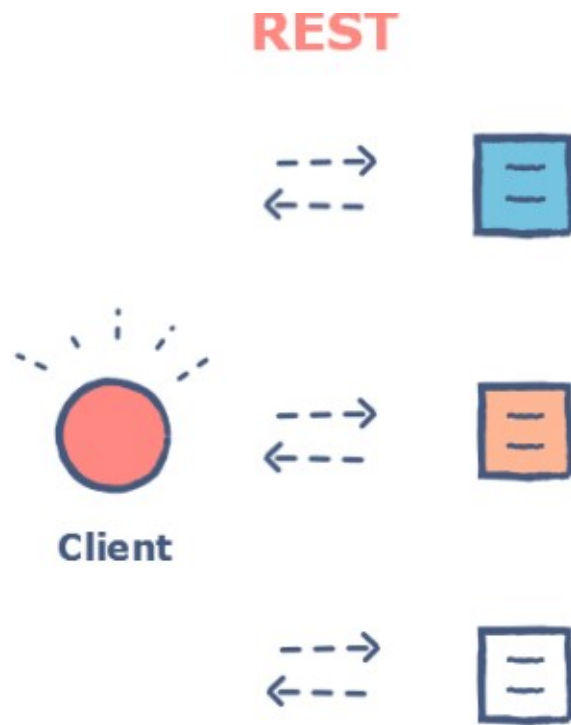
GraphQL

- เนื่องจาก การดึงค่าแบบ **Rest API** บางครั้งข้อมูลมากเกินไปจนจำเป็น บางครั้งต้องดึงหลายครั้ง เพื่อนำข้อมูลมาประมวลผล หรือเอามาประกอบกัน จึงทำให้เกิดแนวคิด ของ **GraphQL**

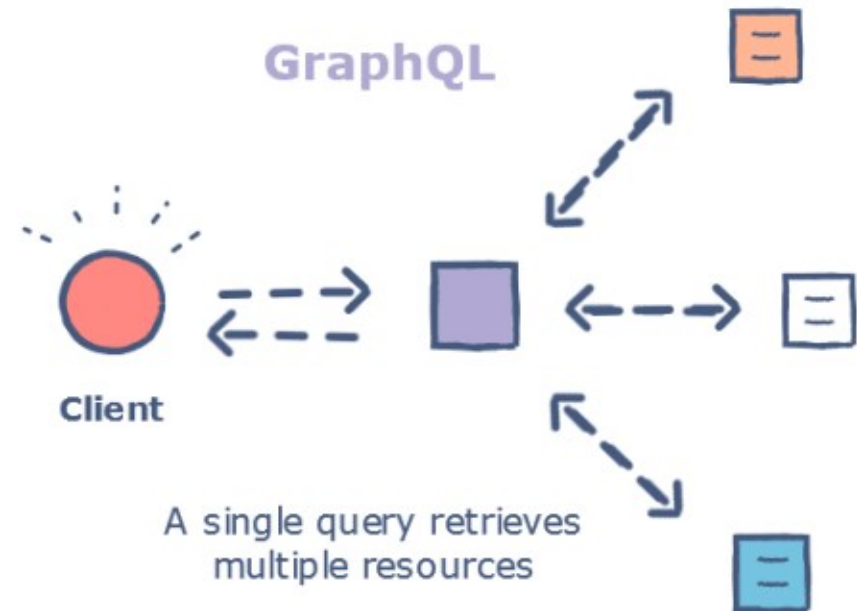
“ GraphQL is a query language for APIs and a runtime for fulfilling those queries with your existing data. ”



GraphQL



Retrieving multiple



ขอบคุณครับ